

## SAFETY DATA SHEET

Creation Date 16-Sep-2010

Revision Date 29-Mar-2024

Revision Number 4

### 1. Identification

**Product Name** Potassium osmium(VI) oxide dihydrate

**Cat No. :** 12647

**CAS No** 10022-66-9

**Synonyms** No information available

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Thermo Fisher Scientific Chemicals, Inc.

30 Bond Street

Ward Hill, MA 01835-8099

Tel: 800-343-0660

Fax: 800-322-4757

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### **Classification**

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                             |            |
|---------------------------------------------|------------|
| Acute oral toxicity                         | Category 3 |
| Acute dermal toxicity                       | Category 3 |
| Acute Inhalation Toxicity - Dusts and Mists | Category 3 |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

Toxic if swallowed, in contact with skin or if inhaled

**Precautionary Statements****Prevention**

Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Wear protective gloves/protective clothing/eye protection/face protection  
Avoid breathing dust/fume/gas/mist/vapors/spray  
Use only outdoors or in a well-ventilated area

**Inhalation**

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing  
Call a POISON CENTER or doctor/physician

**Skin**

IF ON SKIN: Wash with plenty of soap and water  
Call a POISON CENTER or doctor/physician if you feel unwell  
Remove/Take off immediately all contaminated clothing  
Wash contaminated clothing before reuse

**Ingestion**

IF SWALLOWED: Immediately call a POISON CENTER or doctor/physician  
Rinse mouth

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

None identified

### 3. Composition/Information on Ingredients

| Component                                                     | CAS No     | Weight % |
|---------------------------------------------------------------|------------|----------|
| Potassium osmate dihydrate                                    | 10022-66-9 | <=100    |
| Osmate (OsO <sub>4</sub> <sup>2-</sup> ), dipotassium, (T-4)- | 19718-36-6 | -        |

### 4. First-aid measures

**General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

**Ingestion**

Do NOT induce vomiting. Call a physician or poison control center immediately.

|                                            |                                                                                                                                                                                                                                                                                         |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Most important symptoms and effects</b> | None reasonably foreseeable. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                                                                                                                                                                                   |

## 5. Fire-fighting measures

|                                         |                                                                   |
|-----------------------------------------|-------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | CO <sub>2</sub> , dry chemical, dry sand, alcohol-resistant foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                          |
| <b>Flash Point</b>                      | No information available                                          |
| <b>Method -</b>                         | No information available                                          |
| <b>Autoignition Temperature</b>         | No information available                                          |
| <b>Explosion Limits</b>                 |                                                                   |
| <b>Upper</b>                            | No data available                                                 |
| <b>Lower</b>                            | No data available                                                 |
| <b>Oxidizing Properties</b>             | Oxidizer                                                          |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                          |
| <b>Sensitivity to Static Discharge</b>  | No information available                                          |

### Specific Hazards Arising from the Chemical

The product causes burns of eyes, skin and mucous membranes. Oxidizer: Contact with combustible/organic material may cause fire. May ignite combustibles (wood paper, oil, clothing, etc.).

### Hazardous Combustion Products

None known.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

|               |                     |                    |                         |
|---------------|---------------------|--------------------|-------------------------|
| <b>Health</b> | <b>Flammability</b> | <b>Instability</b> | <b>Physical hazards</b> |
| 3             | 1                   | 1                  | OX                      |

## 6. Accidental release measures

|                                             |                                                                                                                                                                                                                                      |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.                                                  |
| <b>Environmental Precautions</b>            | Should not be released into the environment.                                                                                                                                                                                         |
| <b>Methods for Containment and Clean Up</b> | Sweep up and shovel into suitable containers for disposal. Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Sweep up and shovel into suitable containers for disposal. Avoid dust formation. |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                                                                   |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid dust formation. Keep away from clothing and other combustible materials. Use only under a chemical fume hood. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance. |
| <b>Storage.</b> | Keep under nitrogen. Do not store near combustible materials. Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. Store under an inert atmosphere. Protect from moisture. Incompatible Materials. Strong oxidizing agents.                                                                  |

Combustible material. Strong reducing agents. Metals.

## 8. Exposure controls / personal protection

### Exposure Guidelines

This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas.

### Personal Protective Equipment

#### Eye/face Protection

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

#### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure.

#### Respiratory Protection

Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

#### Recommended Filter type:

Particulates filter conforming to EN 143.

#### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                        |                                                       |
|----------------------------------------|-------------------------------------------------------|
| Physical State                         | Solid                                                 |
| Appearance                             | Reddish violet                                        |
| Odor                                   | Odorless                                              |
| Odor Threshold                         | No information available                              |
| pH                                     | Not applicable                                        |
| Melting Point/Range                    | No data available                                     |
| Boiling Point/Range                    | No information available                              |
| Flash Point                            | No information available                              |
| Evaporation Rate                       | Not applicable                                        |
| Flammability (solid,gas)               | No information available                              |
| Flammability or explosive limits       |                                                       |
| Upper                                  | No data available                                     |
| Lower                                  | No data available                                     |
| Vapor Pressure                         | No information available                              |
| Vapor Density                          | Not applicable                                        |
| Specific Gravity                       | No information available                              |
| Solubility                             | Soluble in water                                      |
| Partition coefficient; n-octanol/water | No data available                                     |
| Autoignition Temperature               | No information available                              |
| Decomposition Temperature              | No information available                              |
| Viscosity                              | Not applicable                                        |
| Molecular Formula                      | K <sub>2</sub> O <sub>4</sub> Os . 2 H <sub>2</sub> O |
| Molecular Weight                       | 368.42                                                |

## 10. Stability and reactivity

### Reactive Hazard

Yes

### Stability

Hygroscopic. Oxidizer: Contact with combustible/organic material may cause fire.

|                                         |                                                                                                                 |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Conditions to Avoid</b>              | Incompatible products. Excess heat. Avoid dust formation. Exposure to moist air or water. Combustible material. |
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Combustible material, Strong reducing agents, Metals                                   |
| <b>Hazardous Decomposition Products</b> | None under normal use conditions                                                                                |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                                        |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                                   |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

##### Component Information

| Component                                                     | LD50 Oral                | LD50 Dermal | LC50 Inhalation |
|---------------------------------------------------------------|--------------------------|-------------|-----------------|
| Osmate (OsO <sub>4</sub> <sup>2-</sup> ), dipotassium, (T-4)- | orl-rat LD50: 2694 mg/kg | Not listed  | Not listed      |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** Causes burns by all exposure routes

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                                                     | CAS No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|---------------------------------------------------------------|------------|------------|------------|------------|------------|------------|
| Potassium osmate dihydrate                                    | 10022-66-9 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Osmate (OsO <sub>4</sub> <sup>2-</sup> ), dipotassium, (T-4)- | 19718-36-6 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

|                                      |                                                                          |
|--------------------------------------|--------------------------------------------------------------------------|
| <b>Persistence and Degradability</b> | Soluble in water Persistence is unlikely based on information available. |
| <b>Bioaccumulation/ Accumulation</b> | No information available.                                                |
| <b>Mobility</b>                      | Will likely be mobile in the environment due to its water solubility.    |

### 13. Disposal considerations

|                               |                                                                                                                                                                                                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste Disposal Methods</b> | Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification. |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 14. Transport information

#### DOT

|                             |                                |
|-----------------------------|--------------------------------|
| <b>UN-No</b>                | UN3288                         |
| <b>Proper Shipping Name</b> | TOXIC SOLID, INORGANIC, N.O.S. |
| <b>Technical Name</b>       | Potassium osmate dihydrate     |
| <b>Hazard Class</b>         | 6.1                            |
| <b>Packing Group</b>        | II                             |

#### TDG

|                             |                                |
|-----------------------------|--------------------------------|
| <b>UN-No</b>                | UN3288                         |
| <b>Proper Shipping Name</b> | TOXIC SOLID, INORGANIC, N.O.S. |
| <b>Hazard Class</b>         | 6.1                            |
| <b>Packing Group</b>        | II                             |

#### IATA

|                             |                                |
|-----------------------------|--------------------------------|
| <b>UN-No</b>                | UN3288                         |
| <b>Proper Shipping Name</b> | TOXIC SOLID, INORGANIC, N.O.S. |
| <b>Hazard Class</b>         | 6.1                            |
| <b>Packing Group</b>        | II                             |

#### IMDG/IMO

|                             |                                |
|-----------------------------|--------------------------------|
| <b>UN-No</b>                | UN3288                         |
| <b>Proper Shipping Name</b> | TOXIC SOLID, INORGANIC, N.O.S. |
| <b>Hazard Class</b>         | 6.1                            |
| <b>Packing Group</b>        | II                             |

### 15. Regulatory information

#### United States of America Inventory

| Component                                                     | CAS No     | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|---------------------------------------------------------------|------------|------|-----------------------------------------------|-----------------------------|
| Potassium osmate dihydrate                                    | 10022-66-9 | -    | -                                             | -                           |
| Osmate (OsO <sub>4</sub> <sup>2-</sup> ), dipotassium, (T-4)- | 19718-36-6 | X    | INACTIVE                                      | -                           |

#### **Legend:**

**TSCA** - US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

- - Not Listed

#### **TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

#### **TSCA 12(b) - Notices of Export**

Not applicable

#### **International Inventories**

Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component | CAS No | DSL | NDL | EINECS | PICCS | ENCS | ISHL | AICS | IECSC | KECL |
|-----------|--------|-----|-----|--------|-------|------|------|------|-------|------|
|-----------|--------|-----|-----|--------|-------|------|------|------|-------|------|

|                                                   |            |   |   |           |   |   |  |   |   |          |
|---------------------------------------------------|------------|---|---|-----------|---|---|--|---|---|----------|
| Potassium osmate dihydrate                        | 10022-66-9 | - | - | 243-247-1 | X | - |  | - | - | KE-29169 |
| Osmate (OsO <sub>4</sub> 2-), dipotassium, (T-4)- | 19718-36-6 | - | X | 243-247-1 | - | - |  | - | - | KE-29169 |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### U.S. Federal Regulations

#### **SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372

#### **SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)** Not applicable

**Clean Air Act** Not applicable

**OSHA - Occupational Safety and Health Administration** Not applicable

#### **CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**California Proposition 65** This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations** Not applicable

#### **U.S. Department of Transportation**

Reportable Quantity (RQ): N  
DOT Marine Pollutant N  
DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security** This product does not contain any DHS chemicals.

### Other International Regulations

**Mexico - Grade** No information available

**Authorisation/Restrictions according to EU REACH** Not applicable

| Component                                         | CAS No     | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|---------------------------------------------------|------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Potassium osmate dihydrate                        | 10022-66-9 | -                                                                   | -                                                                             | -                                                                                                     |
| Osmate (OsO <sub>4</sub> 2-), dipotassium, (T-4)- | 19718-36-6 | -                                                                   | -                                                                             | -                                                                                                     |

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                                         | CAS No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|---------------------------------------------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Potassium osmate dihydrate                        | 10022-66-9 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |
| Osmate (OsO <sub>4</sub> 2-), dipotassium, (T-4)- | 19718-36-6 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

## Contains component(s) that meet a 'definition' of per &amp; poly fluoroalkyl substance (PFAS)?

Not applicable

## Other International Regulations

| Component                                         | CAS No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|---------------------------------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Potassium osmate dihydrate                        | 10022-66-9 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Osmate (OsO <sub>4</sub> 2-), dipotassium, (T-4)- | 19718-36-6 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

## Prepared By

Health, Safety and Environmental Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

## Creation Date

16-Sep-2010

## Revision Date

29-Mar-2024

## Print Date

29-Mar-2024

## Revision Summary

New emergency telephone response service provider.

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of SDS